

# Core Programming Foundations



## Python Basics

- Python Intro, Comments, Variables, Data Types
- Numbers, Casting, Strings, Booleans
- Operators, Lists, Tuples, Sets, Dictionaries
- If-Else, While Loop, For Loops
- Functions, Lambda Functions



## Python Advanced

- Arrays, Classes, Objects
- Inheritance, Iterators, Polymorphism
- Scope, Modules, Dates, Math
- JSON, RegEx, PIP
- Try-Except, User Input, File Handling



## Pandas

- Intro, Series, DataFrames
- Read CSV, Read JSON
- Analyze Data, Cleaning Data
- Cleaning Empty Cells, Wrong Format
- Removing Duplicates
- Pandas Correlations, Pandas Plotting

# Database & SQL Mastery



## SQL Fundamentals

- Intro, Syntax, Select, Select Distinct
- Where, Order By, And/Or/Not
- Insert Into, Null Values, Update, Delete
- Select Top, Min/Max, Count/Sum/Avg
- Like, Wildcards, In, Between, Aliases



## SQL Advanced

- Join, Left Join, Right Join, Self Join
- Group By, Having, Exists, Case
- Stored Procedures, Operators
- Create/Drop DB & Table
- Alter DB & Table, Primary/Foreign Key
- Views

# Apache Spark & PySpark Expertise

## PySpark Fundamentals

Features, Advantages, Modules and Packages, ClusterManagers, Installation, Architecture, SparkSession, SparkContext, RDD, Parallelize, Repartition/Coalesce, Broadcast Variables, Accumulator

## PySpark DataFrame Operations

Create Empty DataFrame, Convert RDD to DF, Convert DF to Pandas, Show(), StructType and StructField, Column Class, Select(), Collect(), WithColumn(), withColumnRenamed()

## Advanced DataFrame Methods

where() & filter(), drop() & dropDuplicates(), orderBy() and sort(), groupBy(), join(), union() & unionAll(), unionByName(), UDF, transform(), apply(), map(), flatMap(), foreach()

## PySpark SQL Functions

Aggregate Functions, WindowFunctions, Date and Timestamp Functions, JSON Functions, sample() vs sampleBy(), fillna() & fill(), pivot(), partitionBy(), MapType

# Cloud Platforms: AWS Fundamentals

## AWS Core Services

Intro, Cloud Computing Benefits, EC2, EC2 Instance Types, EC2 Pricing, EC2 Scaling, EC2 Auto Scaling



## Compute & Storage

AWS Load Balancing, AWS Lambda, AWS Containers, AWS Availability Zones, AWS S3, AWS RDS, AWS Redshift



## Management & Security

AWS CloudFormation, AWS Elastic Beanstalk, IAM, CloudWatch, AWS Cloud Compliance

# Databricks: Unified Analytics Platform



## Databricks Basics

Introduction to workspace, clusters, notebooks, DBFS, interactive vs job clusters, connecting to external data sources (S3, ADLS, GCS)



## Data Processing

RDD vs DataFrames Dataset, DataFrame operations, transformations vs actions, joins, aggregations, window functions



## Delta Lake

What is Delta Lake, create/read/write Delta tables, ACID transactions, schema enforcement & evolution, time travel



## ETL Pipelines

Medallion architecture (Bronze <sup>3</sup> Silver <sup>3</sup> Gold), building batch ETL pipelines, handling SCDs, error handling & retries

# Advanced Databricks Features

1

## Data Ingestion & SQL

Reading from CSV, JSON, Parquet; writing to storage (managed vs external tables); Autoloader (cloudFiles); SQL queries and dashboards; basic BI integrations

2

## Job Scheduling & Operations

Scheduling notebooks as jobs, alerts & retries, job runs history, merge operations (UPSERTS), update & delete in Delta, optimize & Z-ORDER, VACUUM & data retention

3

## Streaming & Performance

Read/write streaming data, triggers (microbatch vs continuous), checkpointing, streaming with Delta tables, watermarks & stateful aggregations, partitioning & bucketing, caching & persisting

4

## Advanced Orchestration

Multi-task jobs (DAGs), task dependencies, integrating with Airflow/ADF, parameterized pipelines, Unity Catalog, RBAC, auditing & lineage, Delta Sharing, Delta Live Tables

# Microsoft Azure: Cloud Data Engineering

